

Windows Endpoint Threat Hunting with Sysmon

Incident Investigation Report

Prepared By

Benard Kiprotich

SOC Analyst HomeLab Portfolio Project

Investigation Information

Field	Value
Investigation ID	SOC-2026-003
Investigation Type	Windows Endpoint Threat Hunting
Data Source	Sysmon Operational Log
Primary Event IDs	1 (Process Creation), 11 (File Creation)
Severity	Informational
Status	Closed
Investigation Date	June 2026

Executive Summary

This investigation analyzed endpoint telemetry collected by Sysmon within a Windows 10 virtual machine to identify potentially suspicious process execution and file creation activities. The objective was to understand process relationships, reconstruct user activity and determine whether observed events indicated malicious behavior.

Sysmon Event ID 1 was used to investigate process creation, while Event ID 11 was used to analyze file creation events. Multiple Windows native utilities executed from PowerShell were identified and correlated into a single execution timeline. A scheduled task creation event was also investigated to determine whether it represented a persistence mechanism or legitimate software behavior.

After reviewing all available evidence, the observed activity was assessed as **benign** and consistent with authorized actions performed within a controlled cybersecurity home laboratory.

Investigation Objectives

The objectives of this investigation were to:

- Deploy Sysmon endpoint telemetry for Windows monitoring.
- Investigate Sysmon Process Creation events (Event ID 1).
- Analyze File Creation events (Event ID 11).
- Correlate parent-child process relationships.
- Identify potential indicators of compromise.
- Map observed activity to the MITRE ATT&CK framework.
- Produce a professional SOC incident report.

Lab Environment

Component	Description
Operating System	Windows 10 Virtual Machine
Hypervisor	Oracle VirtualBox
Monitoring Tool	Sysmon
Analysis Tool	Windows Event Viewer
Supporting VM	Kali Linux
Log Source	Sysmon Operational Log

Investigation Timeline

Step	Activity
1	User launched Windows PowerShell from Explorer.
2	PowerShell executed whoami.exe.
3	PowerShell executed hostname.exe.
4	PowerShell executed ipconfig.exe.
5	PowerShell launched Command Prompt (cmd.exe).
6	Command Prompt executed directory listing (dir).
7	Sysmon recorded process creation telemetry.
8	Sysmon recorded a scheduled task file creation event for further investigation.

Evidence Collected

Investigation 1: Process Creation (Sysmon Event ID 1)

Evidence 1

Field	Value
Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Parent Image	C:\Windows\explorer.exe
User	WINDOWS-HOMES-L\vboxuser
Command Line	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

Evidence 2

Field	Value
Image	C:\Windows\System32\whoami.exe
Parent Image	powershell.exe
User	WINDOWS-HOMES-L\vboxuser

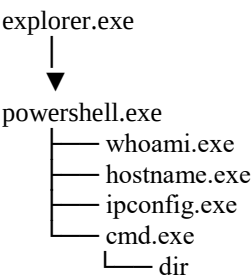
Evidence 3

Field	Value
Image	C:\Windows\System32\ipconfig.exe
Parent Image	powershell.exe
User	WINDOWS-HOMES-L\vboxuser

Evidence 4

Field	Value
Image	C:\Windows\System32\cmd.exe
Parent Image	powershell.exe
User	WINDOWS-HOMES-L\vboxuser

Process Tree



Investigation 2: File Creation (Sysmon Event ID 11)

Evidence

Field	Value
Image	C:\Windows\System32\svchost.exe
Target Filename	C:\Windows\System32\Tasks\GoogleUserPEH
User	NT AUTHORITY\SYSTEM
Process GUID	{17561506-3fee-6a42-1700-000000000c00}
Event	File Creation

Analysis

The investigation identified several legitimate Windows utilities executed from an interactive PowerShell session. The parent-child process relationships showed that PowerShell was launched from Windows Explorer by the logged-on user before spawning `whoami.exe`, `hostname.exe`, `ipconfig.exe`, and `cmd.exe`.

These commands are commonly used for system administration and system information gathering. While they are also frequently observed during the reconnaissance phase of cyber attacks, no suspicious command-line arguments, encoded PowerShell commands or unauthorized parent processes were identified.

A separate Sysmon Event ID 11 recorded the creation of a scheduled task file named **GoogleUserPEH** by `svchost.exe` running as `NT AUTHORITY\SYSTEM`. Although scheduled tasks can be used by attackers to establish persistence, the task name and execution context are consistent with legitimate Google software update operations.

No evidence of malware execution, unauthorized persistence, privilege escalation or command-and-control activity was identified during this investigation.

MITRE ATT&CK Mapping

Technique ID	Technique	Evidence
T1059.001	PowerShell	powershell.exe
T1059.003	Windows Command Shell	cmd.exe
T1033	System Owner/User Discovery	whoami.exe
T1082	System Information Discovery	hostname.exe
T1016	System Network Configuration Discovery	ipconfig.exe
T1053.005	Scheduled Task/Job	Investigation of GoogleUserPEH scheduled task

Indicators of Compromise (IOCs)

No confirmed indicators of compromise were identified.

The observed process execution and scheduled task creation were determined to be consistent with authorized user activity and legitimate system operations.

Conclusion

This project demonstrated the use of Sysmon to investigate endpoint activity through Process Creation and File Creation events. Parent-child process relationships were reconstructed to understand user actions, and a scheduled task creation event was analyzed to determine whether it represented malicious persistence or legitimate software behavior.

The investigation concluded that all observed activity was benign and consistent with authorized operations performed within a controlled cybersecurity home laboratory.

This project demonstrates practical skills in endpoint monitoring, Windows log analysis, Sysmon investigation, threat hunting, MITRE ATT&CK mapping, and professional incident reporting.